Gaming the Two Numbers They Control: Quantization, Weight-Cut Bunching, and the Limits of Strength in 3.9 Million Powerlifting Results

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***Abstract*—A powerlifting competitor controls exactly two numbers: the weight loaded on the bar and their bodyweight at weigh-in. We ask whether these choices leave measurable statistical patterns, using ∼3.94 million competition results from the public-domain OpenPowerlifting database. We report four findings, each answered with a course-appropriate test and led by effect size (at $n \approx 3.9$M almost everything is "significant"). (i) *Quantization*: 96.2% of 13.4M attempt loads sit exactly on the 2.5 kg plate grid; a generalized likelihood-ratio test (GLRT) whose null parameter lies on the boundary is calibrated by a parametric bootstrap. (ii) *Weight-cut bunching*: bodyweight piles up just below every IPF (International Powerlifting Federation) men's class limit (de-heaped $\log(\text{below}/\text{above}) = +1.92$ at 83 kg); a Cattaneo–Jansson–Ma density-discontinuity test rejects at six of the seven limits (surviving Holm and Benjamini–Hochberg); the lone exception is the 120 kg boundary of the open-ended $120+$ class, where the incentive to cut is weakest. A non-limit control shows no bunching, and placebos show no bunching-direction false positives. (iii) *Allometry*: strength scales with bodyweight differently by sex ($b = 0.75$ men, 0.51 women vs. the isometric $2/3$; a Sex$\times \log$(BW) interaction is highly significant and persists on common support). (iv) *Prediction*: bodyweight, sex, equipment, and age predict the total ($R^2 = 0.70$, random forest, leakage-safe grouped CV). Together, the two numbers competitors control leave clear statistical fingerprints, discreteness on the bar and strategic mass below the limit, while the extreme-value analysis finds no evidence of a population strength ceiling. Code and data: https://github.com/adirelm/opl-stat-theory.**

I. Introduction

Powerlifting comprises three lifts (squat, bench press, deadlift), each with three attempts, contested within bodyweight classes. A competitor's *total* is the sum of their best successful attempt in each lift, and is the response variable throughout. Meets differ in permitted supportive gear, from *raw* (belt only) through *single-* and *multi-ply* suits, and in whether the federation is drug-*tested*. Two features make the sport useful for studying strategic behavior. First, the implements are *discrete*: plates come in fixed denominations, so the load a competitor can place on the bar is not a continuous choice. Second, eligibility is governed by a *threshold*: a lifter who weighs even slightly above a class limit is bumped into a heavier, stronger field. We focus on the two numbers a competitor records: the load chosen for each attempt, and the bodyweight at weigh-in. Both interact with a sharp administrative rule. We ask whether these two choices leave detectable statistical signatures, and what ∼3.94M results reveal about the limits of human strength.

Related work. Manipulation of a "running variable" around an administrative threshold produces *bunching*, a standard tool in public economics [1] that is classically tested with the McCrary density-discontinuity test [2] and its modern, bias-corrected local-polynomial form due to Cattaneo, Jansson, and Ma [3]. Rapid pre-weigh-in weight cutting is well documented in weight-class sports [4]. On the physiological side, the square–cube law predicts that strength, governed by muscle cross-sectional area, scales with bodyweight as $\propto \mathrm{BW}^{2/3}$. Extreme-value theory has been used to estimate the ultimate limits of athletic performance from record data [5]. On OpenPowerlifting specifically, recent work studies *scoring and scaling* systems [6].

Contribution. We organize the paper around the two recorded choices, load and weigh-in bodyweight, and use both halves of the course, classical inference and machine learning. Our contributions are: (1) a rounded-likelihood *quantization* GLRT for the plate grid, with a parametric-bootstrap calibration that respects the boundary nature of the null; (2) to our knowledge the first formal density-discontinuity test of weight-cut *bunching* on OpenPowerlifting, checked using a non-limit control, placebo cutoffs, de-heaping, and multiple-testing corrections; (3) a sex-interaction *allometry* model that tests the scaling exponent against the isometric $2/3$; and (4) a leakage-safe strength-*prediction* model.

Methodological stance. At $n \approx 3.9$M the power to detect any non-zero effect is effectively one, so a p-value answers a question we do not care about ("is the effect exactly zero?") rather than the one we do ("is it large?"). We therefore report effect sizes with 95% confidence intervals as the primary evidence, and use p-values only as a secondary screen. H1–H4 are *a-priori* hypotheses, each declared before analysis and tested once; the comparisons that follow an omnibus rejection are *post-hoc* and are the ones that carry a correction. We report all four corrections (Bonferroni, Holm, Šidák, Benjamini–Hochberg) for the H2 family of seven limits, Holm for H3, and Bonferroni for the equipment post-hoc pairs. Because the same lifter appears in many rows, naive standard errors understate uncertainty (pseudo-replication); the formal tests therefore use one row per lifter (with HC3-robust standard errors where the test is regression-based), the deduplication unit declared per

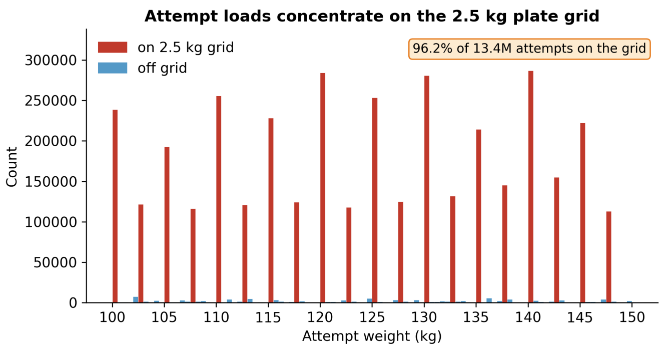

Fig. 1. H1: attempt loads concentrate on the 2.5 kg plate grid (100–150 kg window shown; the 96.2% is computed over all 13.4M attempts).

hypothesis.

II. Results

All intervals are 95%. H1 uses attempt loads; the cross-sectional tests use one row per lifter (to respect independence), and where the class scheme matters the modern kg-only era (2014+). Headline effect sizes for H1 and H2 use all rows; H3's headline exponents are the per-lifter HC3 fits. Throughout, H1–H4 index the four research hypotheses, while H_0 and H_a denote the null and alternative within a given test.

A. Attempt loads are quantized to the 2.5 kg grid (H1)

Of 13,397,702 valid attempt loads, **96.20%** (CI $[96.19, 96.21]$) fall exactly on the 2.5 kg plate grid (Fig. 1); the smallest standard plate is 1.25 kg, loaded on both sides, so 2.5 kg is the minimum increment. To test this formally we restrict to the kg-native (0.5 kg) lattice and model the within-cell position as a mixture that places weight π on the grid point and $1-\pi$ on a uniform background. The GLRT of $\pi = 0$ versus $\pi > 0$ has its null parameter on the boundary of $[0, 1]$, so Wilks' theorem fails: the null distribution of $2 \ln \Lambda$ is the 50:50 mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ of Chernoff and Self–Liang [7], [8]. Our parametric bootstrap closely matches this theory (point mass 0.51 vs. the theoretical 0.50; calibrated 95% cutoff 2.59 vs. the theoretical 2.71, both well below the naive χ^2_1 value of 3.84). The observed statistic, $2 \ln \Lambda = 3.9 \times 10^7$, exceeds every bootstrap draw ($p < 3.4 \times 10^{-4}$), and the Pearson χ^2 goodness-of-fit statistic, the second-order Taylor approximation of the same deviance, agrees (5.0×10^7, df $= 4$). The 3.8% off-grid loads are largely a units artifact: of the 2.1% that fall off the 0.5 kg lattice entirely, 82% are round pound loads (e.g., 102.06 kg= 225 lb), themselves quantized on a different grid.

B. Bodyweight bunches below class limits (H2)

Bodyweight piles up just below real class limits (Fig. 2). On de-heaped data (exact round/half-kg weigh-ins removed, so the signal is not mere digit preference), the log-ratio of just-below to just-above counts is positive at all seven IPF men's limits and strongest at 83 kg ($+1.92 \pm 0.04$, a $6.8\times$ asymmetry; Table I, Fig. 3); a non-limit control at 91 kg is flat-to-slightly-negative (-0.21). The formal Cattaneo–Jansson–Ma density-discontinuity test [3] (one random meet per lifter, de-heaped, 2014+) rejects at six of the seven limits with the density dropping at the cutoff ($t < 0$); the lone exception is the 120 kg cutoff ($t = -1.6$, $p = 0.11$), adjacent to the unbounded 120+ class where the incentive to cut is weakest. The control shows no excess below (a small *opposite*-direction discontinuity), and no placebo shows the bunching pattern. The effect is robust across equipment (raw $+2.14$, equipped $+1.51$) and concentrated in the tested majority ($+1.93$; the smaller untested subset reverses to -0.30, consistent with weight-cutting being driven by the tested IPF-scheme federations that dominate this population), and it persists on the clean kg-only subset ($+1.80$).

TABLE I
De-heaped $\log(\text{below}/\text{above})$ at each IPF men's limit (all rows; counts over the 0.5 kg windows just below/above each limit). Last row is a non-limit control.

Limit (kg)	log-ratio	$\pm$	$\times$ asymmetry
59	$+0.99$	0.04	2.7
66	$+1.18$	0.03	3.2
74	$+1.00$	0.03	2.7
83	$\mathbf{+1.92}$	0.04	**6.8**
93	$+1.65$	0.04	5.2
105	$+1.17$	0.05	3.2
120	$+0.74$	0.06	2.1
91 (control)	-0.21	0.03	0.8

Robustness and falsification. The result survives four checks. (i) *De-heaping*: removing exact round/half-kg weigh-ins leaves the asymmetry intact, so the signal is not digit preference and is consistent with weight-cut manipulation. (ii) *Control*: the non-limit point at 91 kg shows no excess below. (iii) *Placebos*: cutoffs placed between real limits show no bunching-direction discontinuity after correction (two show opposite-sign jumps, density rising rather than excess mass below, like the control). (iv) *Subgroups and correction*: the effect holds within raw, equipped, and tested subsets, and the formal density test rejects at six of the seven limits even under the strictest (Bonferroni) correction.

C. Allometric scaling differs by sex (H3)

Fitting $\log(\text{Total}) = a + b \log(\text{BW})$ (Fig. 4), the exponent is $b = 0.75$ (CI $[0.74, 0.75]$) for men and 0.51 ($[0.50, 0.51]$) for women, with heteroskedasticity-robust (HC3) standard errors on one row per lifter ($n = 545{,}728$ men, 224,458 women). A two-sided Wald test rejects the isometric $b = 2/3$ for both sexes, and in *opposite* directions (men $z = +46.8$; women $z = -64.1$). A pooled Sex$\times \log(\text{BW})$ interaction estimates the men-minus-women slope gap at $+0.24$ ($p \ll 10^{-3}$). The gap also *grows* on the common bodyweight support (0.84 men vs. 0.38 women), so it is not an artifact of the sexes occupying different weight ranges. For men, the Pearson correlation of $\log \text{BW}$ and $\log \text{Total}$ is 0.60 (Fisher-transform CI $[0.600, 0.603]$; Spearman 0.58); it is lower for women.

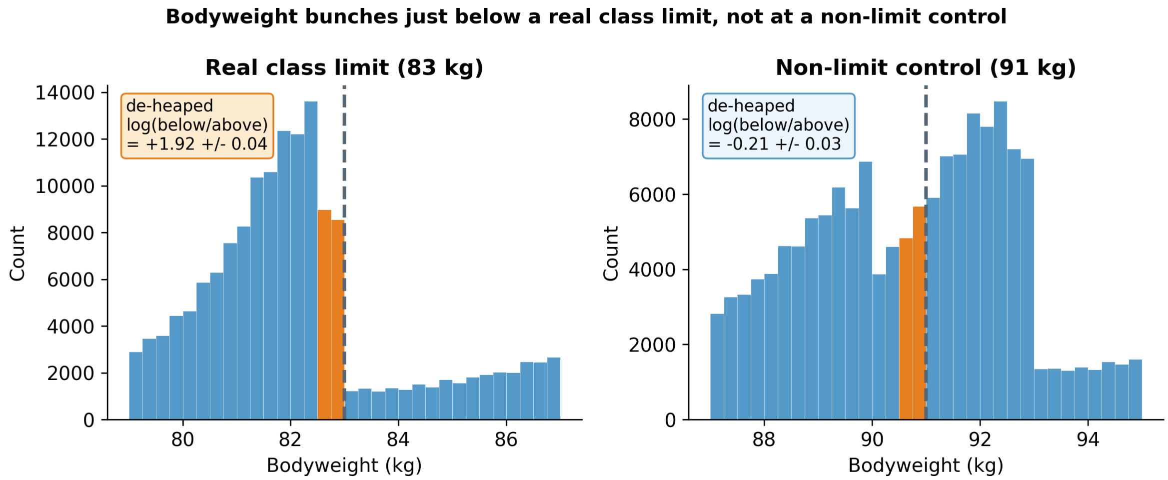

Fig. 2. H2: bunching just below a real limit (83 kg) but not at a non-limit control (91 kg); round-number weigh-ins removed.

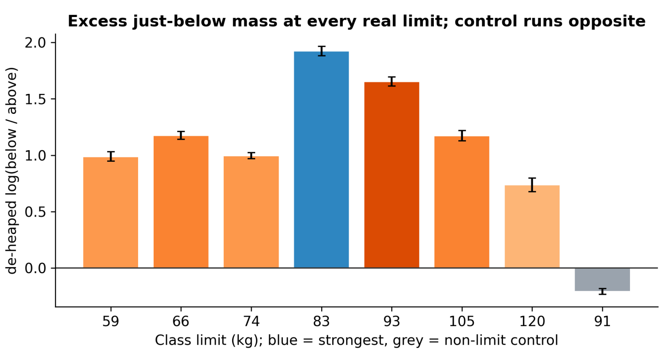

Fig. 3. H2: excess just-below mass at every real limit; the control runs opposite.

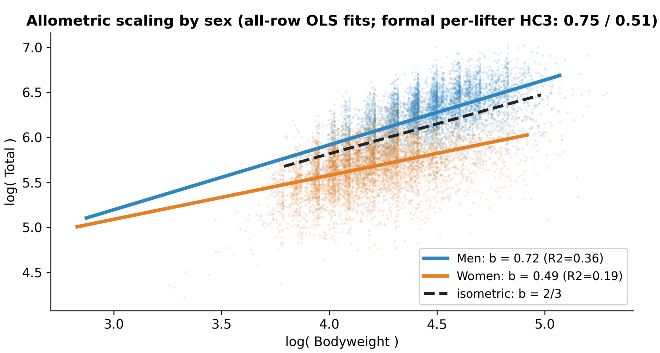

Fig. 4. H3: allometric scaling by sex vs. the isometric $2/3$. Plotted lines are all-row OLS fits; the headline per-lifter HC3 exponents are 0.75 (men) / 0.51 (women).

The all-row log-log fit explains $R^2 = 0.36$ of the variance in men and 0.19 in women. The message is not that women are weaker, but that the *estimated scaling* differs; we leave the mechanism to future work.

D. Strength is predictable from basic variables (H4)

Predicting the total from controllable/basic variables (bodyweight, sex, equipment, age), on one random meet per lifter subsampled to a seeded $200{,}000$ lifters for tractability, with every split grouped by lifter and all preprocessing fit inside each training fold, a random forest reaches $R^2 = 0.70$ (RMSE 86 kg) versus 0.61 (RMSE 98 kg) for linear regression (Fig. 5); permutation importance is dominated by sex and bodyweight, and continuous-feature VIFs are ≈ 1 (no collinearity concern). A logistic "made-weight" classifier built from *non-bodyweight* features (age, tested status, era, equipment; the Dots score, a bodyweight- and sex-adjusted strength index, is excluded because it is a function of total and bodyweight) reaches $\text{AUC} = 0.56$ on $n = 89{,}882$ lifters (PR-AUC 0.84 against a 0.82 no-skill baseline, i.e. the below-limit share; balanced accuracy 0.53; at the 0.5 threshold precision 0.85 and recall 0.38). Adjusted R^2 equals R^2 to two decimals for the linear model (0.61 with five features), so its fit is not an artifact of feature count. The signal is weak, as expected: cutting is driven by bodyweight relative to the limit, which defines the label and which these features omit.

E. Supporting analyses

Distribution structure. The apparent two-component shape of raw totals is just sex: by BIC and a bootstrap likelihood-ratio test, raw Total prefers two Gaussian components ($\Delta\text{BIC} = 1.4 \times 10^3$) but the sex-normalized Dots score prefers one (Fig. 6). **Tested vs. untested.** Untested competitions show higher Dots than tested ones (medians 344 vs. 318, Mann–Whitney $p \to 0$; a parametric Welch test agrees, $|t| = 128$; rank-biserial -0.20, point-biserial -0.14), consistent with (though not proof of) an effect of performance-enhancing drugs, since tested and untested federations also differ in era, equipment, and selection. **Paired attempts.** Within a meet, the third squat attempt exceeds the opener by a median 15 kg (medians $160 \to 180$; $n = 10^5$ pairs; Wilcoxon signed-rank $p \to 0$), the paired-sample counterpart

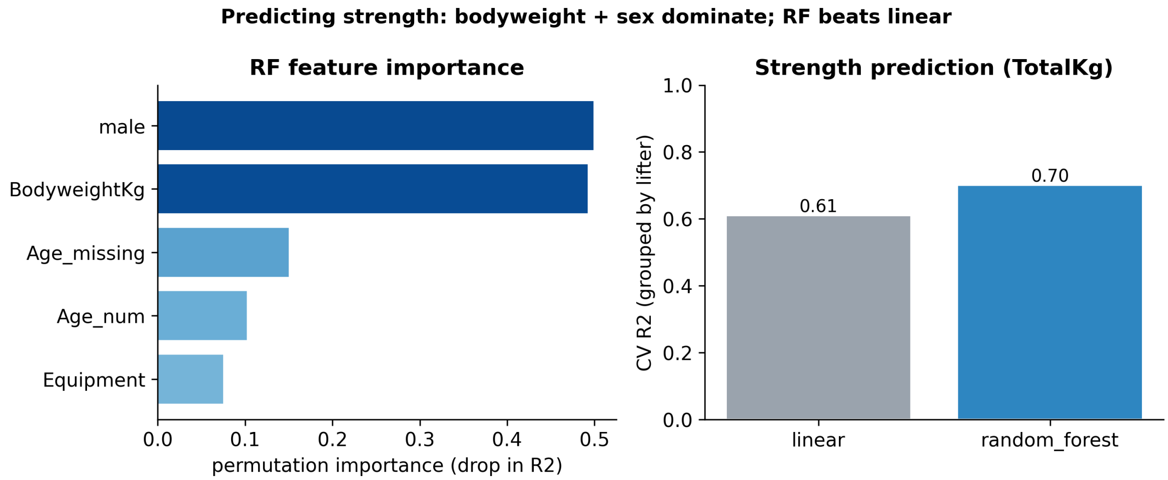

Fig. 5. H4: strength prediction; importance and grouped-CV R^2.

of the two-sample comparison above. Ties are pervasive here by construction (H1), so the normal approximation carries the continuity correction and the 2,570 zero differences are dropped. Scored made/missed, the same pairs give a McNemar $\chi^2 = 1.4 \times 10^4$ (df = 1): third attempts fail far more often than openers. **Equipment.** Dots differs across equipment categories (ANOVA $F = 5.3 \times 10^3$; Kruskal–Wallis agrees, $H = 1.2 \times 10^4$; all six Bonferroni post-hoc pairs reject), with multi-ply highest (median Dots 401 vs. 328 raw). Group variances are unequal (Levene $W = 3.4 \times 10^3$), so we also report Welch's ANOVA ($F = 3.6 \times 10^3$), which agrees; the effect is nonetheless small ($\eta^2 = 0.02$). **Extreme values.** A GEV fit to 1,514 equal-size blocks of 250 per-lifter totals (men, 2015+) gives shape $\xi = -0.02$ with bootstrap CI $[-0.06, 0.01]$: the shape is indistinguishable from zero, so a bounded upper tail (a population strength *ceiling*) is *not established*. Seeding the GEV fit from its moment match is what makes this interval meaningful; with the default start a sixth of the bootstrap fits diverge and inflate the upper limit. **Time trend.** Median Dots drifted down over 2000–2024 (Spearman $\rho = -0.54$); the annual series is strongly autocorrelated (lag-1 residual ACF 0.73, Ljung–Box $p < 10^{-3}$), so we read ρ descriptively. One possible explanation is broadening participation, but the trend is not by itself evidence of declining strength. **Sequential detection and power.** A Wald sequential probability ratio test on the chronological stream of just-below/above indicators accepts the bunching hypothesis after a stopping time of only $N = 27$ (Fig. 7). An a-priori calculation for the same design ($p = 0.6$ vs. 0.5) requires $n = 211$ for 0.90 power at $\alpha = 0.05$, and makes the Type I/II tradeoff concrete: at fixed $n = 100$, tightening α from 0.10 to 0.01 raises β from 0.23 to 0.63. **Independence.** The association between tested status and cutting is negligible either way (Cramér's $V < 0.01$); removing pseudo-replication flips the nominal significance (all rows $p = 0.02$; one row per lifter $\chi^2 = 3.3$, $p = 0.07$, matched by Fisher's exact $p = 0.09$), a reminder that at scale significance tracks repeated measurements, not effect size. No cell drives the table (largest adjusted standardized residual $|z| = 1.8 < 1.96$; we use the row/column-adjusted form because only it is standard normal under independence, and for a 2×2 table Cramér's V coincides with ϕ). The untested arm holds only 30 lifters (9 above, 21 below), so this is an underpowered non-result rather than evidence of no association. **Residual normality.** The H3 (all-row fit) residuals are left-skewed with a heavy lower tail (skew -0.77; Fig. 8). This does not undermine the slope: at this n its sampling distribution is asymptotically normal under standard conditions, and we report HC3-robust standard errors. At $n = 300$ all three normality tests the course names agree in rejecting (Shapiro–Wilk $W = 0.96$, $p = 8 \times 10^{-7}$; Anderson–Darling $A^2 = 2.55$ against a 0.75 critical value; Kolmogorov–Smirnov $D = 0.09$, $p = 0.01$), as expected for genuinely non-normal residuals, so we treat normality as a diagnostic and lead with effect sizes.

III. Methods

Data and preprocessing. We use a pinned snapshot of OpenPowerlifting [9] (3,941,811 rows, public domain; the SHA-256 is recorded for exact reproducibility). The analysis samples are reached in three steps: 3,941,811 raw entries $\to$ 2,627,657 with a valid full-power total and bodyweight $\to$ 770,186 after collapsing to one row per lifter (198,740 for the 2014+ kg-only H2 subset). The unit of analysis is the *attempt* for H1 (the mechanism is within-competitor, so the grid is robust to clustering; attempts do cluster within lifters, so the attempt-level CI is optimistic in width and the conclusion rests on the effect size, 96% on-grid vs. the 20% no-preference baseline) and one *row per lifter* for the cross-sectional tests; *de-heaping* drops exact round/half-kg weigh-ins so the bunching signal is not digit preference. For prediction, all cross-validation is grouped by lifter so the same athlete never appears in both train and test.

GLRT and the boundary (H1). With log-likelihood ℓ, the statistic is $2 \ln \Lambda = 2(\ell_1 - \ell_0)$. Under regularity $2 \ln \Lambda \to \chi^2_r$, but when the null fixes a parameter on the boundary of its

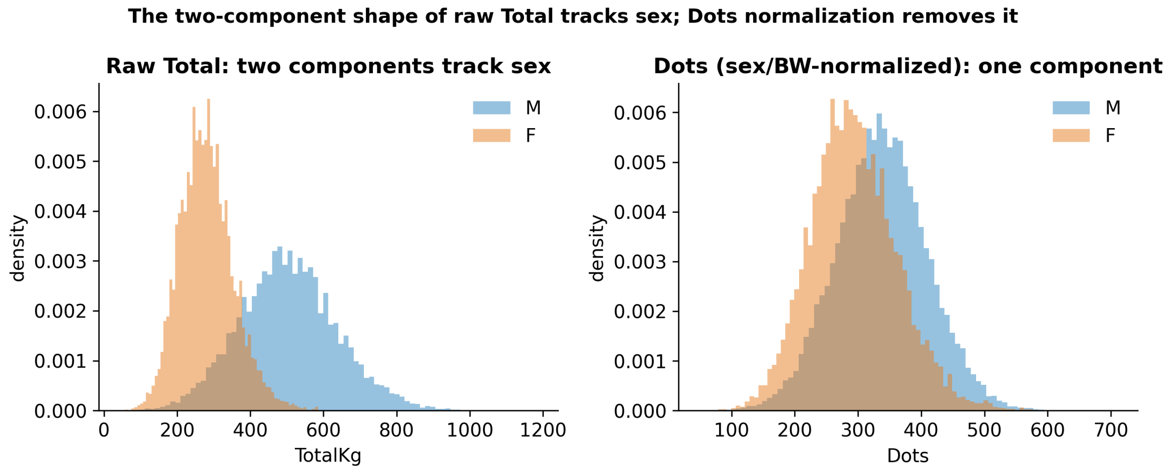

Fig. 6. The apparent "mixture" in raw Total is sex; Dots removes it.

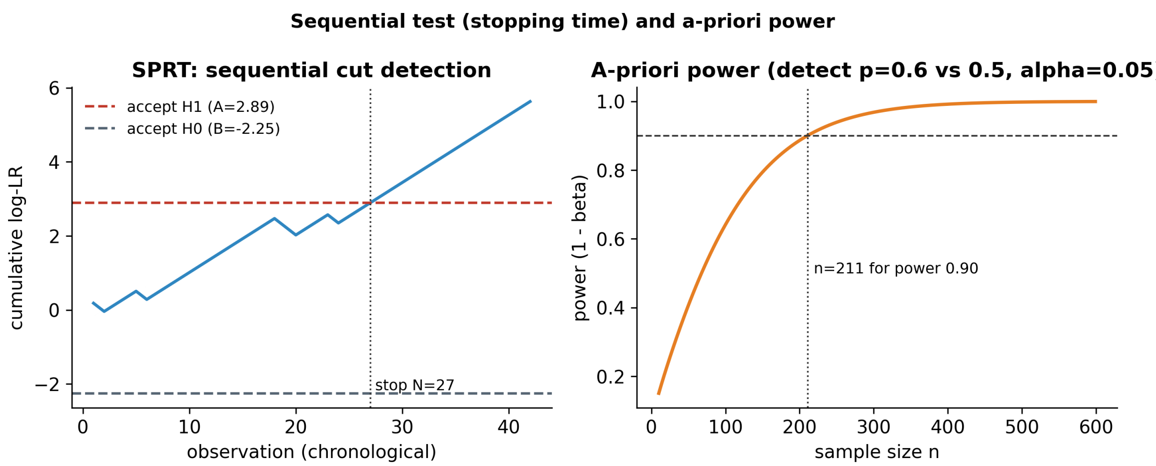

Fig. 7. Sequential test (stopping time $N = 27$) and a-priori power ($n = 211$ for power 0.90).

space the limit becomes a mixture $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ [7], [8]; we estimate the null distribution directly by simulating data under the fitted null model (parametric bootstrap) and comparing the observed statistic to the simulated quantiles. Concretely, for within-cell position $u \in \{0, \ldots, 4\}$ on the 0.5 kg lattice the model is

$$P(u) = \pi\, \mathbf{1}[u = 0] + (1 - \pi)\tfrac{1}{5}, \qquad \pi \in [0, 1], \tag{1}$$

and the GLRT contrasts $\pi = 0$ (no grid preference, on the boundary) with the constrained MLE $\hat{\pi} \geq 0$.

Density discontinuity (H2). For a cutoff c the manipulation test contrasts the one-sided densities, $\theta = \ln \hat{f}^{+}(c) - \ln \hat{f}^{-}(c)$, estimated by local polynomials without pre-binning [2], [3]; $\theta < 0$ ($t < 0$) indicates excess mass just below c. We complement the formal test with a de-heaped log count-ratio as the interpretable effect size, counting weigh-ins over the windows $(c - 0.5, c]$ and $(c, c + 0.5]$ kg. A bare count ratio presumes rather than tests a discontinuity and depends on the window, which is why the formal decision rests on the local-polynomial density test.

Allometry (H3). OLS on log–log data gives the scaling exponent b; we report HC3-robust standard errors (one row per lifter), a Wald statistic $W = (\hat{b} - 2/3)/\widehat{\mathrm{se}}(\hat{b})$, and a pooled Sex$\times$ log(BW) interaction whose coefficient is exactly the men-minus-women slope difference. Correlations use the Fisher z-transform for confidence intervals.

Learning and the rest of the toolbox. Prediction uses linear and random-forest regression and logistic classification, evaluated by grouped cross-validation (R^2, adjusted R^2, RMSE, MAE, permutation importance, VIF, AUC/PR-AUC). The 1-versus-2-component mixture LRT violates Wilks for the same reason as H1 (the mixing weight lies on the boundary and the second component is unidentified under the null), so its null is bootstrapped rather than read off a χ^2; and no course tool addresses a bounded tail, since the sample maximum is not asymptotically normal, which is why the ceiling question is posed in extreme-value terms. Supporting analyses use a bootstrap mixture-LRT with AIC/BIC, a GEV extreme-value fit, a Wald sequential probability ratio test

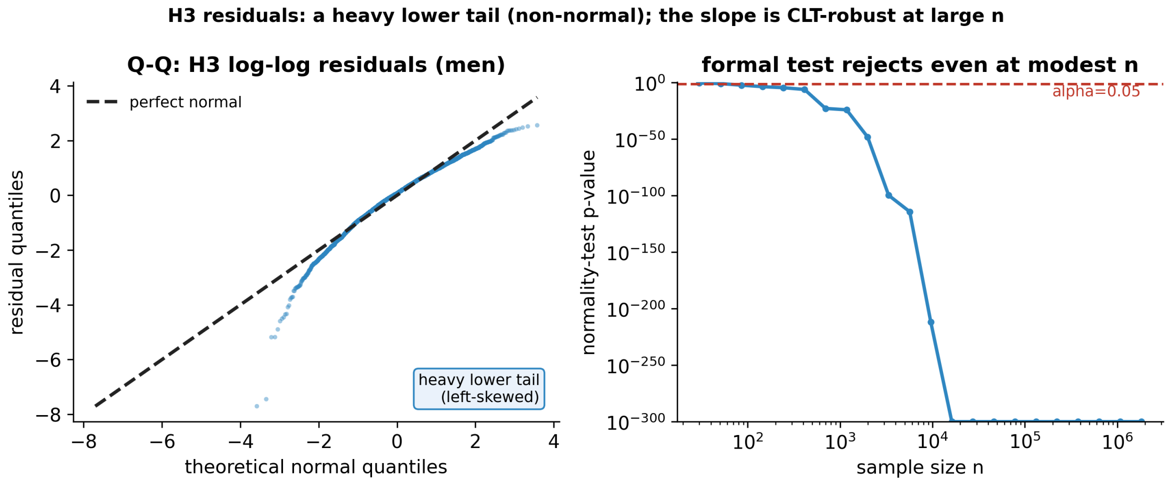

Fig. 8. H3 residuals: heavy lower tail; the slope is CLT-robust at large n.

(with log-scale decision boundaries $A = \ln\frac{1-\beta}{\alpha}$ to reject H_0 and $B = \ln\frac{\beta}{1-\alpha}$ to accept it, a stopping time, and an expected-sample-size approximation), one- and two-sample nonparametric tests (Wilcoxon signed-rank; Mann–Whitney), an F-test/Kruskal–Wallis omnibus with post-hoc comparisons, and a χ^2 independence test with Cramér's V and standardized residuals. The one-sided test of $b > 2/3$ for men ($z = 46.8$) is our Neyman–Pearson example: uniformly most powerful in the one-parameter exponential-family idealization, with the GLRT as its composite-hypothesis generalization. The extreme-value fit uses the GEV distribution

$$G(x) = \exp\Big\{-\big(1+\xi\tfrac{x-\mu}{\sigma}\big)^{-1/\xi}\Big\}, \tag{2}$$

where $\xi < 0$ implies a bounded upper tail (a strength ceiling) and $\xi \geq 0$ does not; we estimate ξ from equal-size block maxima and bootstrap its interval. The sequential test's expected sample size under the alternative is approximated by $\mathbb{E}[N \mid H_1] \approx \big[(1-\beta)A + \beta B\big]/\big[p_1 s_1 + (1-p_1)s_0\big]$, with s_0, s_1 the per-observation log-likelihood increments. Table II maps the course toolbox onto the study.

Reproducibility. Every result is produced by a small set of scripts from the pinned snapshot; numbers are written to machine-readable files, figures are exported at 300 DPI, and an independent re-derivation from the raw data reproduced every headline value reported here. All analyses use Python 3.12 with numpy/scipy/statsmodels/scikit-learn and the `rddensity` implementation of the CJM test; exact versions are pinned in the repository.

TABLE II
COVERAGE OF THE COURSE TOOLBOX.

Concept	Where used
GLRT, MLE, χ^2 goodness-of-fit	H1 grid model
Boundary LRT, parametric bootstrap	H1 null calibration
Density discontinuity (McCrary/CJM)	H2 bunching
Multiple testing (Bonf./Holm/Šidák/BH)	H2 limits, H3
OLS, R^2, Wald, interaction term	H3 allometry
One-/two-sided, one-sample tests	H1, H3
Neyman–Pearson MP/UMP framing	H3 one-sided test
Pearson/Spearman/point-biserial, Fisher CI	H3, tested vs. untested
Regression, random forest, logistic	H4 prediction
Learning metrics, grouped CV, VIF	H4
Mixture LRT, AIC/BIC	structure
Welch t / Mann–Whitney (two-sample)	tested vs. untested
Wilcoxon signed-rank (paired)	attempt 1 vs. 3
McNemar (paired categorical)	attempt made/missed
F/ANOVA, Kruskal–Wallis, post-hoc	equipment
χ^2 independence, Cramér's V/ϕ	tested × cut
Levene, Welch ANOVA, η^2	equipment variances
Shapiro–Wilk / A–D / K–S	H3 residual normality
Extreme-value theory (GEV)	strength ceiling
Sequential Wald (SPRT), stopping time	cut detection
Power, Type I/II tradeoff	a-priori power

IV. DISCUSSION

Conclusions. The two numbers a competitor controls leave measurable, falsification-surviving traces. Attempt loads are quantized to the plate grid; bodyweight bunches specifically below class limits, and not at controls or placebos in the bunching direction. Beyond strategy, the data also bear on the limits of strength: scaling with bodyweight is sex-specific and departs from the isometric $2/3$, and basic meet-record variables already explain $\sim$70% of the variance in the total.

Limitations and null results. This is an observational study: we identify population-level patterns consistent with weight cutting, not the intent of any individual. We report three null results as genuine findings. The 120 kg boundary (the entry to the unbounded 120+ class) does *not* show significant bunching after correction ($t = -1.6$, $p = 0.11$), consistent with weaker cutting pressure at that open-ended boundary; the extreme-value analysis does *not* support a strength ceiling (the ξ confidence interval includes zero); and the tested-versus-cutting comparison is under-powered in this IPF-restricted population (only 30 untested lifters near a limit), so it remains open rather than settled. On the methods side, lifter identity is name-based, the per-lifter meet is sampled at random, and the formal bunching test is restricted to the modern kg-only era. The 83 kg signal is robust to the federation set: the deheaped log-ratio is $+1.92$ across the IPF-affiliated federations used here and $+1.66$ when restricted to IPF/USAPL alone. Several extensions remain for future work: more federations and eras, a structural bunching model that recovers the share of manipulators, and a multiverse robustness check.

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